

FUNCTIONAL AREAS OF CEREBRAL CORTEX

DR. Fatima Haq

Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

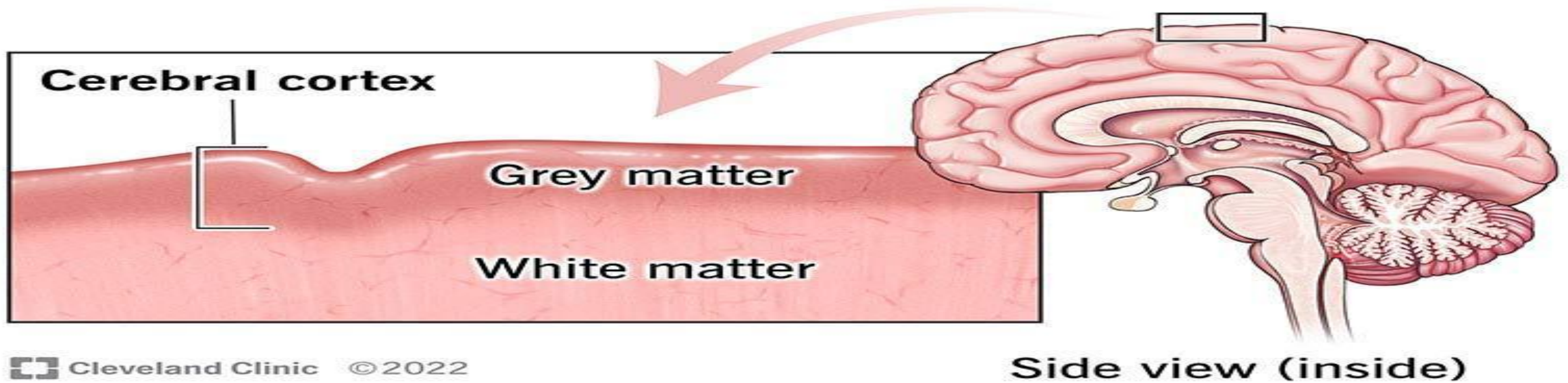
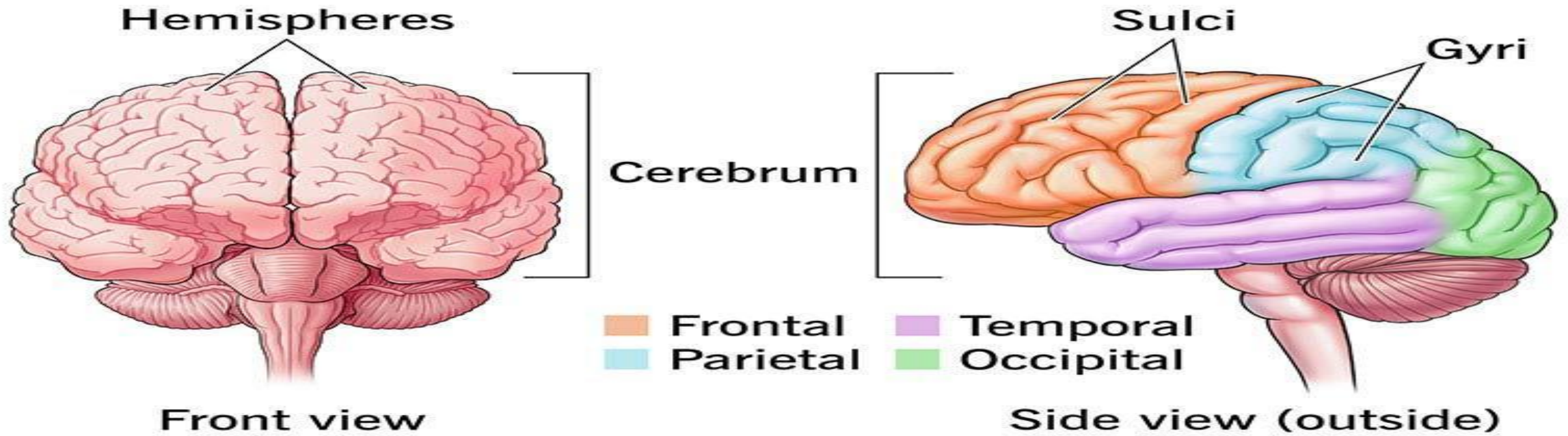
By the end of this lecture students should be able to:

- ✓ Identify major functional areas of cerebral cortex & their locations in frontal, parietal, temporal & occipital lobes.
- ✓ Differentiate between primary, secondary & association areas of cerebral cortex.
- ✓ Describe the functions of primary motor, premotor, prefrontal & sensory cortical areas.
- ✓ Explain the functions of primary somatosensory, visual & auditory cortices.
- ✓ Identify the cortical areas involved in language.

What is cerebral cortex?

- Cerebral cortex is the ***outer layer of gray matter*** covering the cerebral hemispheres.
- It contains billions of neurons organized into functional networks.
- Phylogenetically, it is divided into ***neocortex & allocortex***.
- The cerebral cortex is composed of ***six cellular*** layers from outside to inside namely,
 1. Molecular (plexiform) layer
 2. External granular layer
 3. Outer pyramidal layer
 4. Internal granular layer
 5. Ganglionic (internal pyramidal) layer
 6. Fusiform layer

Cerebral cortex

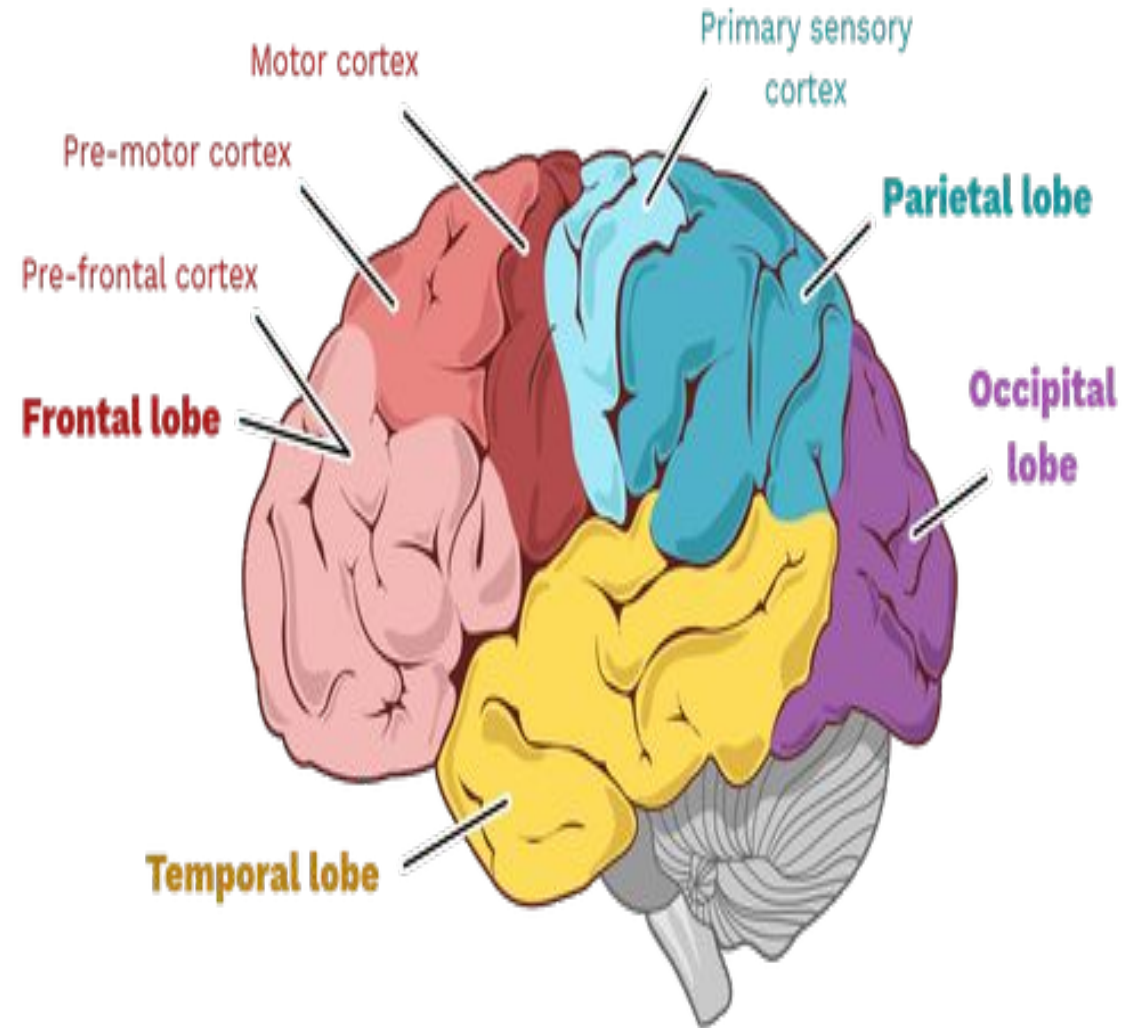


FUNCTIONS OF CEREBRAL CORTEX

□ The cortex is responsible for:

1. Conscious perception
2. Voluntary movement
3. Language
4. Memory
5. Thinking
6. Learning
7. Planning
8. Personality & behaviour

□ The cortex does not function as isolated areas rather different cortical regions are communicated through extensive neural networks.



FUNCTIONAL AREAS OF CEREBRAL CORTEX

□ Functionally, the cerebral cortex can be divided into:

1. **Primary areas:** perform basic sensory or motor functions e.g. primary motor cortex, primary somatosensory cortex, primary visual cortex & primary auditory cortex.
2. **Secondary areas:** interpret information from a particular sensory or motor modality e.g. premotor cortex, supplementary motor area, secondary somatosensory cortex, secondary visual cortex & secondary auditory cortex.
3. **Association areas:** integrate information from multiple sensory modalities & involved in higher mental functions.

Functional Areas of the Cerebral Cortex

Motor areas

Primary motor cortex

Premotor cortex

Frontal
eye field

Broca's area
(outlined by dashes)

Prefrontal cortex

Central sulcus

Sensory areas and related association areas

Primary somatosensory
cortex

Somatosensory
association cortex

Somatic
sensation

Gustatory cortex
(in insula on roof
of lateral sulcus)

Taste

Wernicke's area
(outlined by dashes)

Primary visual
cortex

Visual
association
area

Vision

Auditory
association area

Primary
auditory cortex

Hearing

(a) Lateral view, left cerebral hemisphere



Primary motor cortex



Motor association cortex



Primary sensory cortex



Sensory association cortex



Multimodal association cortex

LOBES OF CEREBRAL CORTEX

□ Each cerebral hemisphere consists of four lobes i.e.

1. ***Frontal lobe***
2. ***Parietal lobe***
3. ***Occipital lobe***
4. ***Temporal lobe***

□ Lobes of each hemisphere are demarcated by four main fissures or sulci namely:

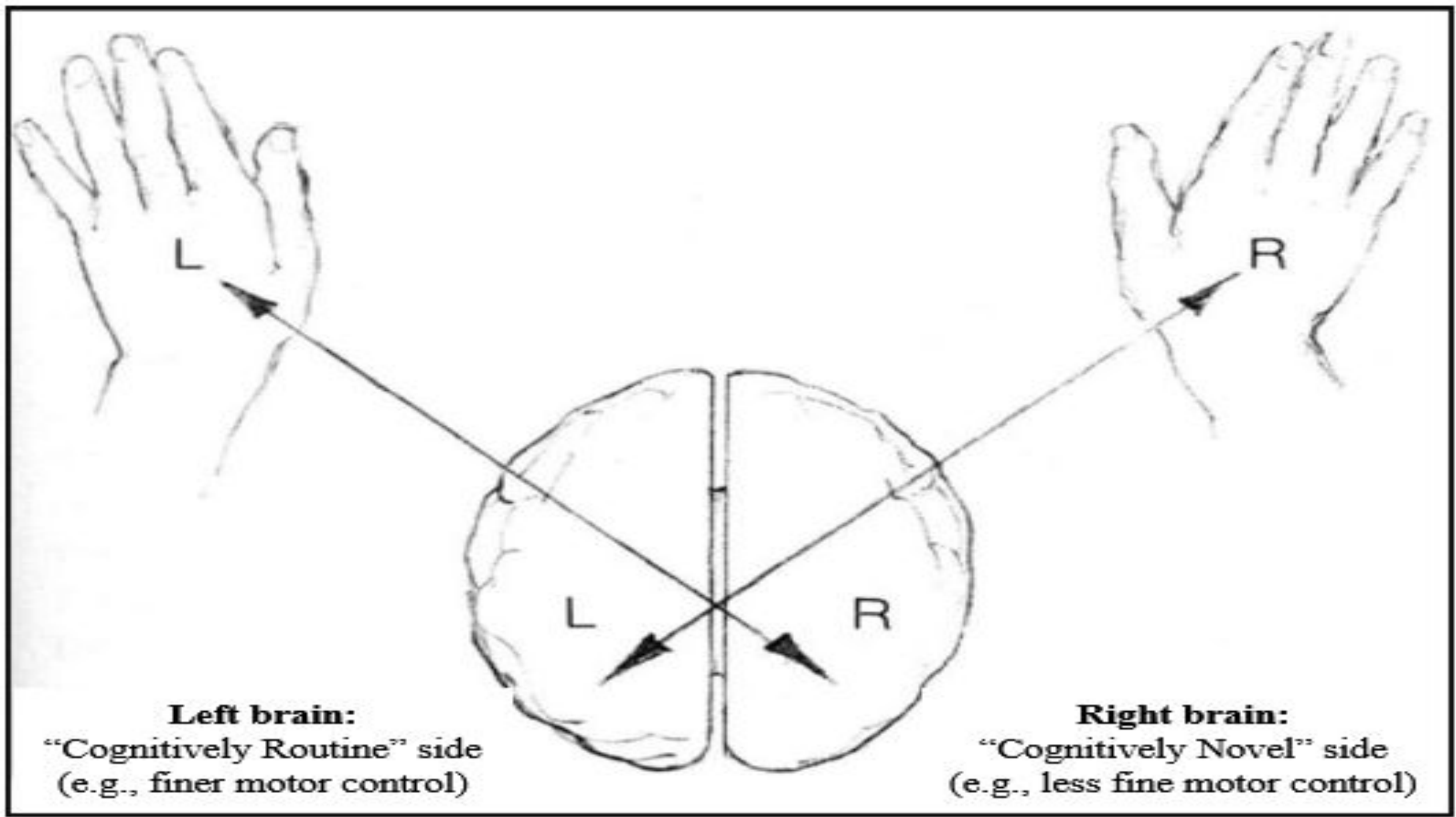
- i. ***Central sulcus or Rolandic fissure***: between frontal & parietal lobes.
- ii. ***Parieto-occipital sulcus***: between parietal & occipital lobes.
- iii. ***Sylvian fissure or lateral sulcus***: between parietal & temporal lobes
- iv. ***Callosomarginal fissure***: between temporal lobe & limbic area.

What is cerebral dominance?

- It is defined as the dominance of one cerebral hemisphere over the other in the control of cerebral functions.
- Both the cerebral hemispheres are not functionally equivalent.

CEREBRAL DOMINANCE & HANDEDNESS

- Cerebral dominance is related to handedness i.e. preference of the individual to use right or left hand.
- More than 90% of people are right-handed i.e. the left hemisphere is ***dominant or categorical hemisphere*** & controls analytical process & language related functions like speech, reading & writing.
- Right hemisphere in these individuals is the ***representational hemisphere*** & associated with artistic & visuospatial functions like judging the distance, determining the direction, recognizing the tones etc.



FRONTAL LOBE OF CEREBRAL CORTEX

- Frontal lobe forms one third of the cortical surface.
- It extends from frontal pole to the central sulcus & limited below by the lateral sulcus.
- Frontal lobe is divided into:
 - I. Precentral cortex
 - II. Prefrontal cortex

PRECENTRAL CORTEX

- It is also called excitomotor cortex or area as the stimulation of different points in this area causes activity of discrete skeletal muscles.
- It forms the posterior part of frontal lobe & includes the lip of central sulcus, whole of precentral gyrus & posterior portions of superior, middle & inferior frontal gyri.

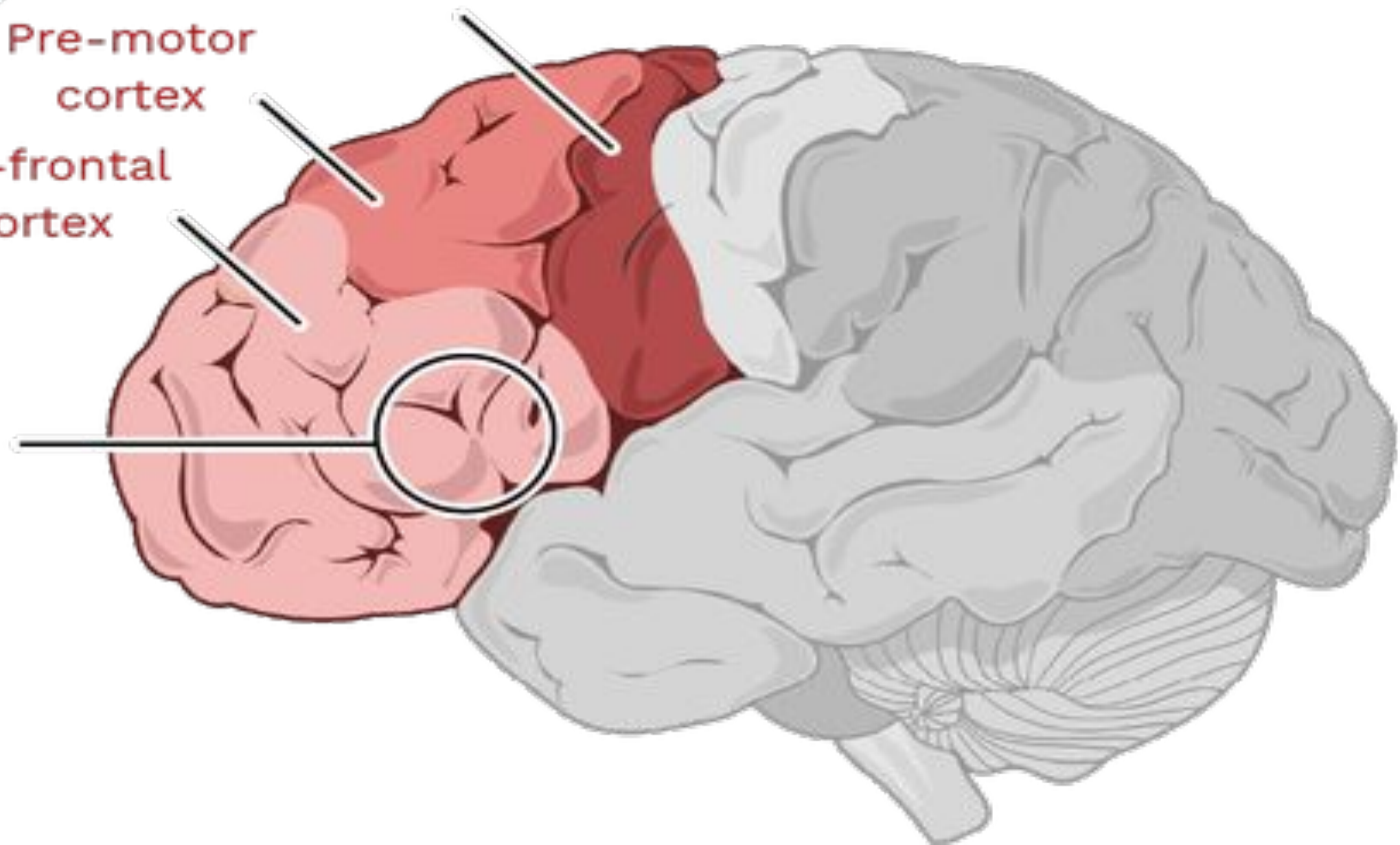
**Frontal
lobe**

Motor cortex

Pre-motor
cortex

Pre-frontal
cortex

Broca's
area



Cont..

- Precentral cortex is divided into three functional areas:
 - I. Primary motor area
 - II. Premotor area
 - III. Supplementary motor area

PRIMARY MOTOR AREA

- It extends throughout the precentral gyrus & the adjoining lip of central sulcus.
- Areas 4 & 4S are present in primary motor area.
- Special structural feature of this layer is the presence of giant pyramidal cells called ***Betz cells*** in the ganglionic layer.

AREA 4

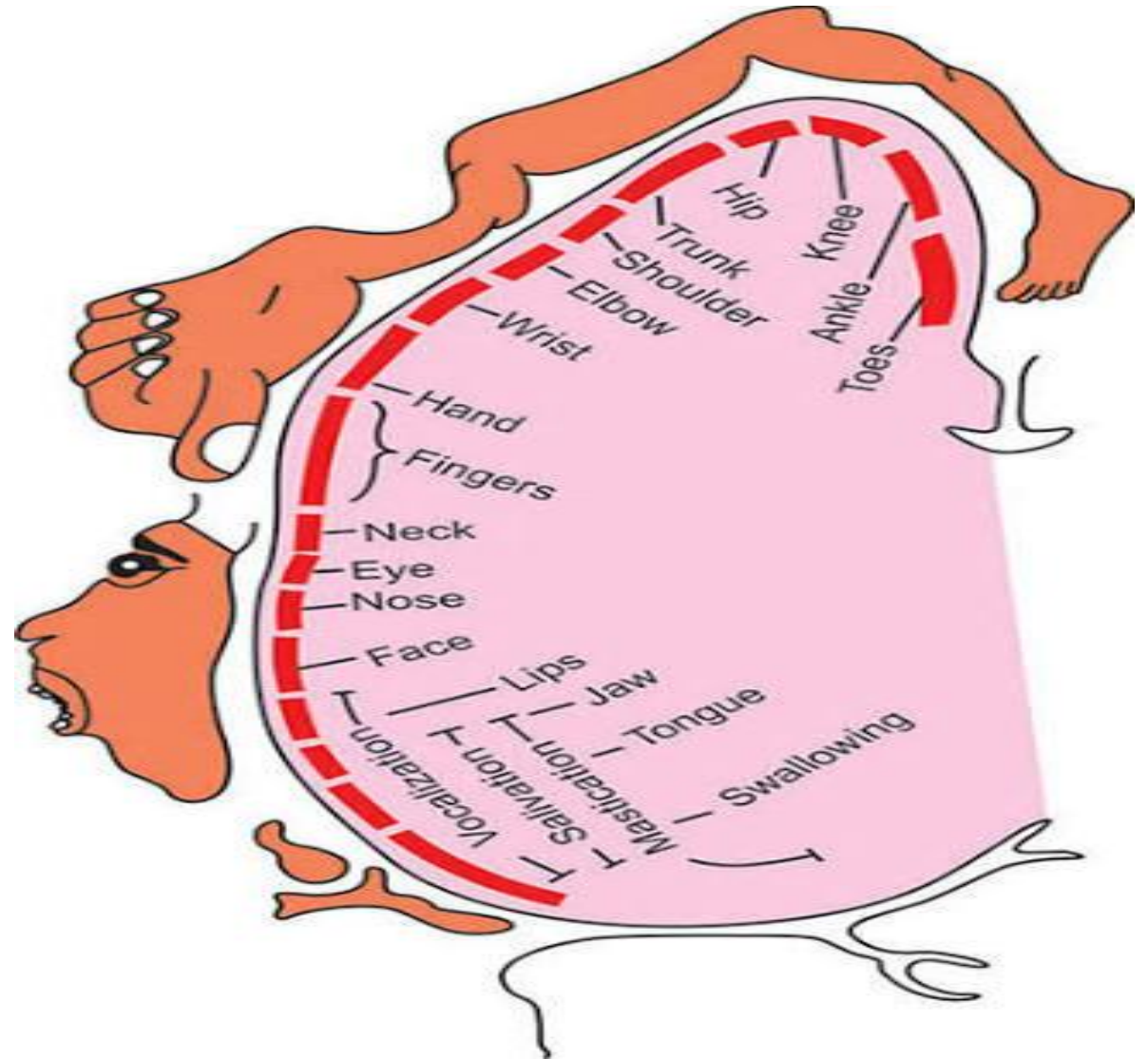
- Tapering strip of area situated in precentral gyrus of frontal lobe.
- Broad end lies superiorly at the upper border of hemisphere from where most of the efferent fibers of primary motor area originate.
- It is the ***center for movement*** as it sends corticospinal fibers of primary motor area.

FUNCTIONS OF AREA 4

- Through corticospinal tract, area 4 activates the lower motor neurons in the spinal cord.
- It activates both alpha motor neurons & gamma motor neurons simultaneously(coactivation).
- Activation of alpha motor neuron causes contraction of ***extrafusal fibers*** of the muscles whereas activation of gamma motor neurons causes contraction of ***intrafusal fibers*** which increase muscle tone.

HOMUNCULUS OF MOTOR AREAS

- Muscles of various parts of body are represented in area 4 in an inverted way from medial to lateral surface.
- Lower parts of the body are represented in medial surface while upper parts are represented in the lateral surface.



Cont..

- Area 4 sends motor signals to the facial muscles of both sides(bilateral) & other muscles of opposite side(contralateral).
- If area 4 is affected along with area 6, it causes **hemiplegia**(paralysis in one half of body) with **spastic paralysis**(spastic contraction of muscles due to increased muscle tone).

AREA 4S

- It is the **suppressor area** & comprise of a narrow strip anterior to area 4.
- It suppresses the extra impulses produced by area thus inhibiting exaggeration of movements.

PREMOTOR AREA

- It lies anterior to primary motor area in the precentral cortex.
- It includes area 6, 8, 44 & 45.

AREA 6

- Area 6 lies in the posterior portions of superior, middle & inferior frontal gyri.
- It is sub-divided into **6a** & **6b**.
- Stimulation of area 6a causes rotation of head, eyes & trunk towards the opposite side.
- Stimulation of area 6b produces rhythmic movements involving muscles of face, buccal cavity, pharynx & larynx.

FUNCTIONS OF AREA 6

- Area 6 is the cortical center for extra-pyramidal system.
- It is concerned with the coordination of movements initiated by area 4 & helps to make the skilled movements more accurate & smooth.

AREA 8 (FRONTAL EYE FIELD)

- It lies anterior to area 6 in the precentral cortex.
- It sends efferent fibers to oculomotor nuclei in the midbrain & is associated with movements of the eyeball.

FUNCTIONS OF AREA 8

- It is concerned with ***conjugate movement of eyeballs*** to the opposite side.
- It is also responsible for opening & closing of eyelids, pupillary dilatation & lacrimation.

BROCA'S AREA(AREA 44 & 45)

- Broca's area is the motor area for speech.
- It is situated in the inferior frontal gyrus of dominant hemisphere.
- Area 44 lies in ***pars triangularis*** of inferior frontal gyrus & area 45 is situated in ***pars opercularis*** of this gyrus.

FUNCTION OF BROCA'S AREA

- Broca's area is responsible for movements of structures involved in speech i.e.tongue, lips & larynx.

SUPPLEMENTARY MOTOR AREA(SMA)

- It is situated in medial surface of frontal lobe rostral to primary motor area.
- It is associated with the coordination of skilled movements.
- Motor movements like raising contralateral arm, turning the head & eye & movements of synergistic muscles of trunk & legs are caused by the stimulation of this area.

PREFRONTAL CORTEX (ORBITOFRONTAL CORTEX)

- It is the anterior part of frontal lobe & situated in front of area 8 & 44.
- It includes orbital gyri, medial frontal gyrus & the anterior portions of superior, middle & inferior frontal gyri.
- Areas 9, 10, 11, 12, 13, 14, 23, 24, 29 & 32 are present in prefrontal cortex.

FUNCTIONS OF PREFRONTAL CORTEX

- It is the ***center for higher functions*** like emotion, learning, memory & social behaviour.
- ***Short term memories*** are stored in this region.
- It is the ***center for planned actions***.
- It is also the ***seat of intelligence*** hence called the ***organ of mind***.
- This area is responsible for the personality of individual.
- It also causes various autonomic changes(like changes in the activity of digestive,cardiovascular,respiratory & excretory systems) during emotional conditions.

FRONTAL LOBE FUNCTIONS:

- Coordinating voluntary movements
- Impulse control (decision making)
- Assessing similarities and differences between two objects
- Forming and retaining long-term memories
- Understanding Language, linguistic memories, and speaking
- Emotional regulation and expression
- Personality development
- Attention regulation



INJURIES TO THE FRONTAL LOBE MAY CAUSE:

- inability to engage in decision-making or poor decision-making skills
- Difficulties with planning, executive functioning, and attention
- Loss of memory
- Sudden or dramatic changes in personality
- Changes in emotions
- Difficulty understanding social cues
- Difficulty with empathy or relating to other's emotions
- Changes in motor skills and spatial reasoning abilities

PARIETAL LOBE OF CEREBRAL CORTEX

- It extends from central sulcus & merges with occipital lobe behind & temporal lobe below.
- Parietal lobe is separated from occipital lobe by parieto-occipital sulcus & from temporal lobe by sylvian sulcus.
- It is divided into three functional areas:
 - i. Somesthetic area I
 - ii. Somesthetic area II
 - iii. Somesthetic association area
- In addition to these areas, a part of sensory motor area is also situated in parietal lobe.

Brain Lobes Functions

Frontal Lobe

- Executive function
- Planning & decision-making
- Emotional regulation

Parietal Lobe

- Somatosensory processing
- Spatial awareness
- Sensory integration

Occipital Lobe

- Visual processing
- Pattern recognition

Temporal Lobe

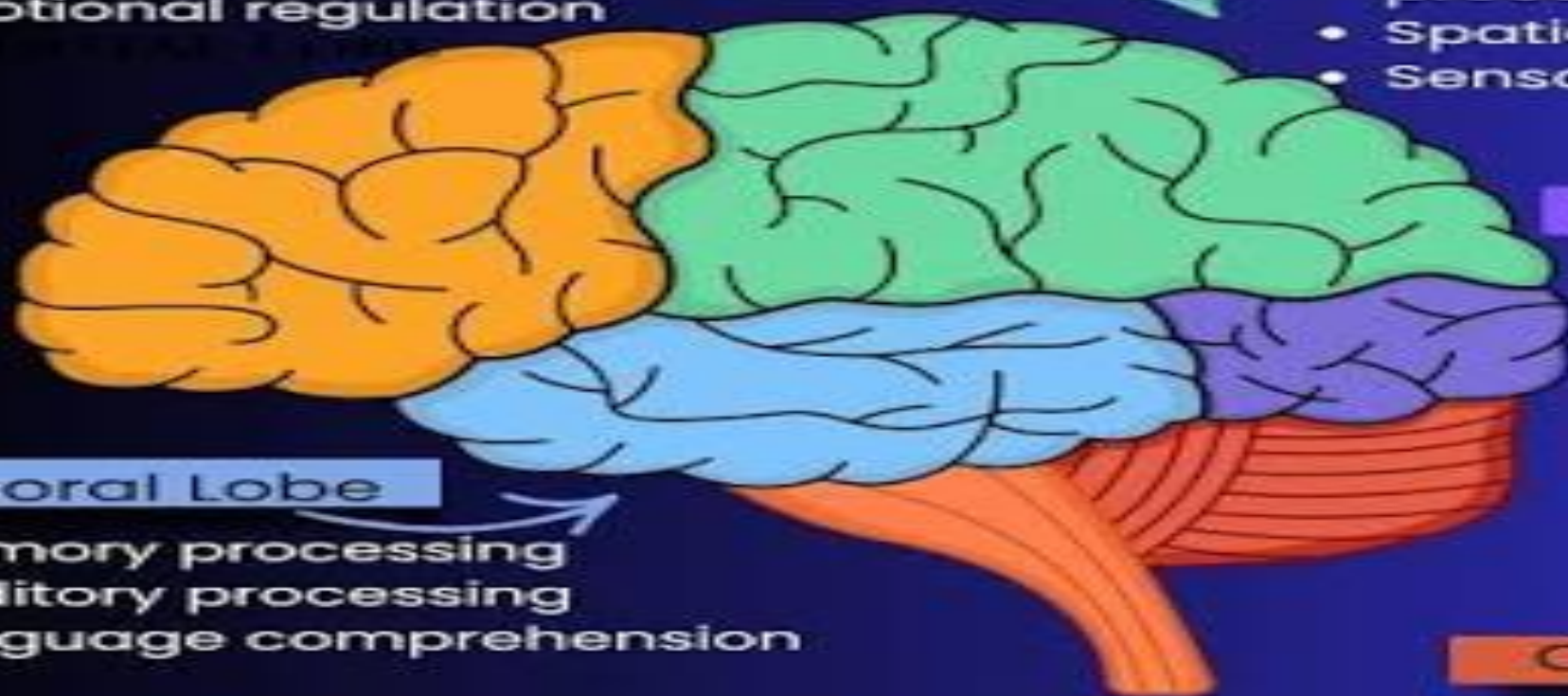
- Memory processing
- Auditory processing
- Language comprehension

Brain Stem

- Breathing
- Heart rate regulation
- Alertness
- Digestion

Cerebellum

- Coordination
- Balance
- Motor control



SENSORY MOTOR AREA

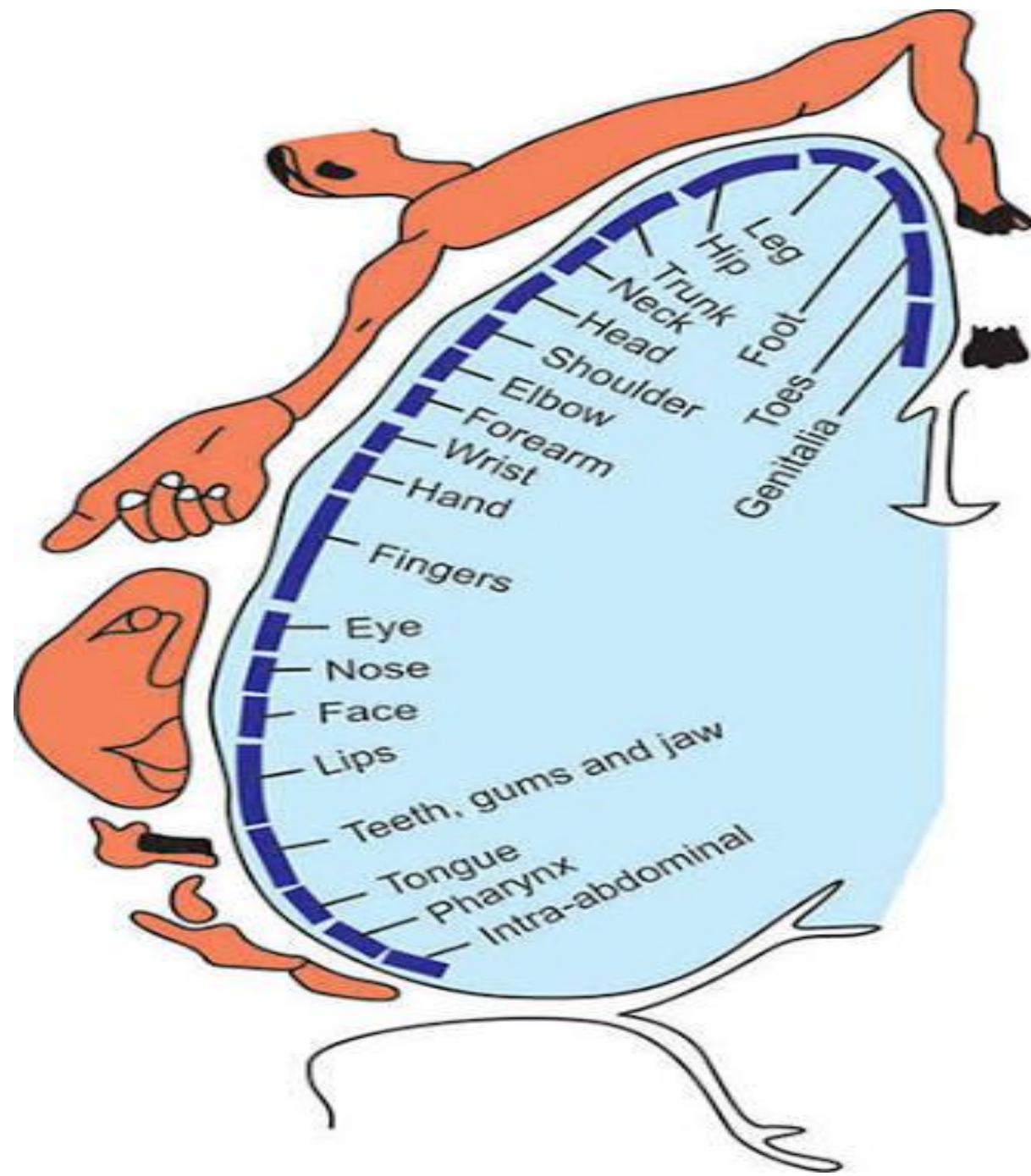
- Sensory area of cortex is not limited to the postcentral gyrus in parietal lobe, it extends anteriorly into motor area situated in precentral gyrus of frontal lobe.
- Similarly, motor area is extended posteriorly from precentral gyrus into the postcentral gyrus of parietal lobe.
- The precentral & postcentral gyri are inter-related & connected together by association neurons.
- So, this area is termed as sensory motor area.

SOMESTHETIC AREA I

- It is also called ***somatosensory area I*** or primary somesthetic or ***primary sensory area***.
- It is situated in the posterior lip of central sulcus, in the postcentral gyrus.
- Somesthetic area I has three areas i.e. area 3, 1 & 2.
- Anterior part of somesthetic area I forms area 3 & posterior part forms area 1 & 2.

HOMUNCULUS OF SENSORY AREAS

- Different sensory areas of the body are represented in the post-central gyrus of primary sensory area in an inverted manner.
- Representation of face is not inverted & parts of face from above downwards is eyelids, nose, cheek, upper lip & lower lip.



FUNCTIONS OF SOMESTHETIC AREA I

- Somesthetic area I is responsible for perception & integration of cutaneous & kinesthetic sensations.
- It receives sensory impulses from cutaneous receptors(touch, pressure, pain, temperature)& proprioceptors of opposite side.
- Area 1 is concerned with sensory perception whereas areas 3 & 2 are involved in the integration of these sensations.
- Somesthetic area I sends sensory feedback to the premotor area.
- In addition to perception of cutaneous & kinesthetic sensation, somesthetic area I is also responsible for recognizing the discriminative features of sensations.
- Stimulation of somesthetic area I produces numbness & tingling sensation.

SOMESTHETIC AREA II

- It is also called ***secondary somesthetic area*** or ***somatosensory area II***.
- It is situated in the postcentral gyrus & a part of somesthetic area II is buried in sylvian sulcus.

FUNCTIONS OF SOMESTHETIC AREA II

- It receives sensory impulses from somesthetic area I & from thalamus directly.
- Although the exact role of this area is unclear, it is involved in the perception of sensation.

SOMESTHETIC ASSOCIATION AREA

- It is situated posterior to postcentral gyrus above the auditory cortex & in front of visual cortex.
- Somesthetic association area has two areas i.e. area 5 & 7.

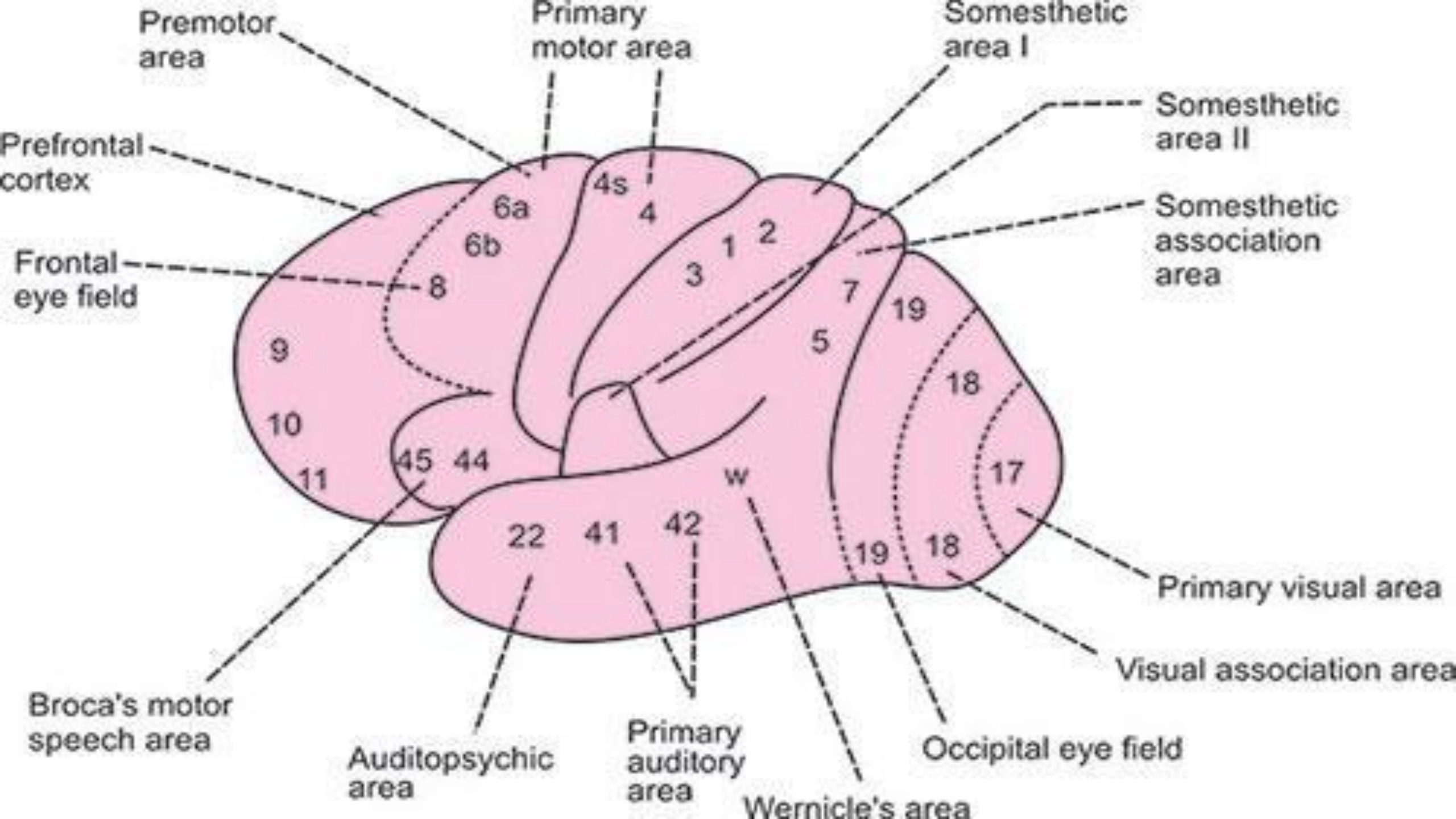
FUNCTIONS OF SOMESTHETIC ASSOCIATION AREA

- Somesthetic association area forms the center of stereognosis(i.e. identify shape, size , texture of object using only the sense of touch).
- It integrates tactile & proprioceptive data to interpret object's shape, size, texture & body position.

Parietal Lobe: Main Functions

- Spatial awareness & navigation
- Sensory integration (touch, pain, pressure, temperature)
- Body perception & movement coordination
- Reading, writing & mathematical abilities
- Attention allocation
- Perception of environment & object manipulation





TEMPORAL LOBE OF CEREBRAL CORTEX

□ It includes three functional areas:

- i. Primary auditory area
- ii. Secondary auditory area
- iii. Area for equilibrium

PRIMARY AUDITORY AREA

□ It includes area 41, 42 & Wernicke's area.

□ Area 41 & 42 are situated in the anterior transverse gyrus & lateral surface of superior temporal gyrus.

□ Wernicke's area is located in the upper part of superior temporal gyrus posterior to areas 41 & 42.

FUNCTIONS OF PRIMARY AUDITORY AREA

□ Area 41 & 42 are involved in the perception of sound(i.e. analysis of pitch & determination of intensity & source of sound).

Cont..

- Wernicke's area is responsible for the interpretation of auditory sensation(sound).
- Wernicke's area is also responsible for understanding the auditory information of words & sending this information to broca's area.

SECONDARY AUDITORY AREA

- It is also called as ***auditopsychic area*** or ***auditory association area*** & includes area 22.
- It occupies the superior temporal gyrus.
- It is concerned with the interpretation of auditory sensation along with Wernicke's area.
- It is also important for the storage of memories of spoken words.

Temporal Lobe

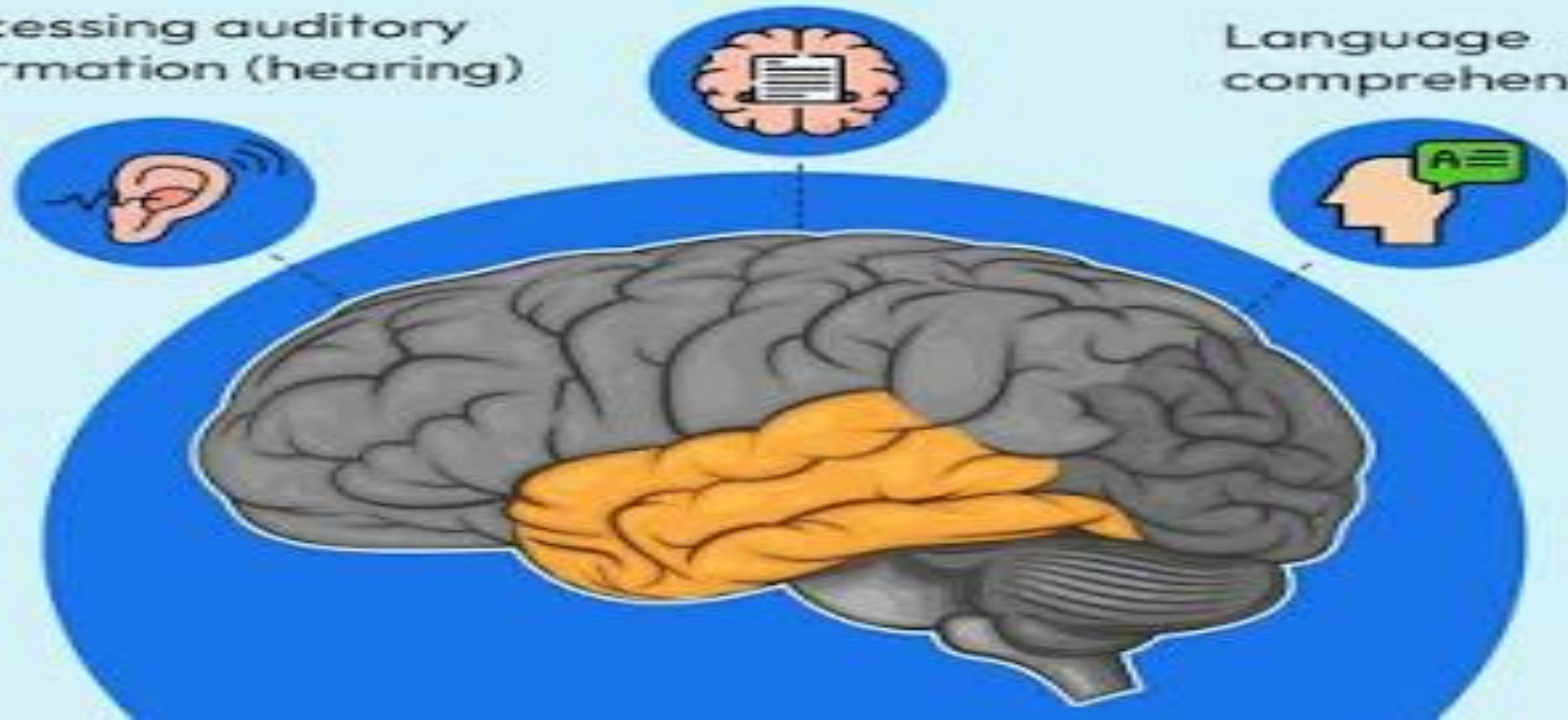
The temporal lobe is located near the temples, on the lower sides of the brain.

Its main functions include:

Memory formation
and learning

Processing auditory
information (hearing)

Language
comprehension



AREA FOR EQUILIBRIUM

- It is located in the posterior part of superior temporal gyrus & involved in maintaining the equilibrium of body.

OCCIPITAL LOBE OF CEREBRAL CORTEX

- It is also termed as ***visual cortex***.
- It consists of three functional areas:
 - i. Primary visual area(area 17)
 - ii. Secondary visual area(area 18)
 - iii. Occipital eye field(area 19)

FUNCTIONS OF OCCIPITAL LOBE

- Area 17 is concerned with the perception of visual sensation (sight).
- Area 18 is associated with the interpretation of visual sensation & storage of memories of visual symbols.
- Area 19 is concerned with the reflex movement of eyeballs & associated movements of eyeballs while following a moving object.

The Occipital Lobe

is located at the back of our brain and is divided into two hemispheres, the left and the right.

It is responsible for processing and interpreting visual information received from our eyes. Imagine it as the **“visual processor”** of our brain.

Occipital Lobe



	Brodmann's Area	Name	Function	Connections
Frontal lobe:	4	Primary motor cortex	Voluntary muscle activation	Contributes to corticospinal tract
	6	Premotor cortex		
	8	Frontal eye field	Eye movements	Sends projections to lateral gaze center (paramedian pontine reticular formation)
	44, 45	Broca's area	Motor aspects of speech	Projects to Wernicke's area via arcuate fasciculus
Parietal lobe:	3, 1, 2	Primary sensory cortex	Somatosensory	Input from VPL, VPM
Occipital lobe:	17	Striate cortex = primary visual cortex	Processing of visual stimuli	Input from lateral geniculate only Projects to areas 18, 19
	18, 19	Extrastriate = visual association cortex	Processing of visual stimuli	Input from area 17
Temporal lobe:	41	Primary auditory cortex	Processing of auditory stimuli	Input from medial geniculate
	42	Associative auditory cortex		
	22	Wernicke's area	Language comprehension	Inputs from auditory association cortex, visual association cortex, Broca's area (via arcuate fasciculus)

POST ASSESSMENT

Q1. A patient develops weakness of the right hand & face following a cortical lesion. Which area is most likely affected?

- a) Right primary motor area
- b) Left primary motor area
- c) Left primary visual area
- d) Right somatosensory association cortex

Q2. A patient can hear sounds but has difficulty interpreting spoken words. Which cortical area is most likely affected?

- a) Broca's area
- b) Wernicke's area
- c) Premotor cortex
- d) Primary motor area

Cont..

Q3. Primary visual cortex is located mainly in the:

- a) Frontal lobe
- b) Parietal lobe
- c) Temporal lobe
- d) Occipital lobe

Q4. Which cortical area is primarily involved in planning & programming voluntary movements?

- a) Premotor area
- b) Primary motor area
- c) Somatosensory cortex
- d) Wernicke's area